

Phase 7: Integration & External Access – Donation Tracking System for NGO's

- **Step 1: Named Credentials (Store API login securely)**

1. Go to Setup → Named Credentials → New.

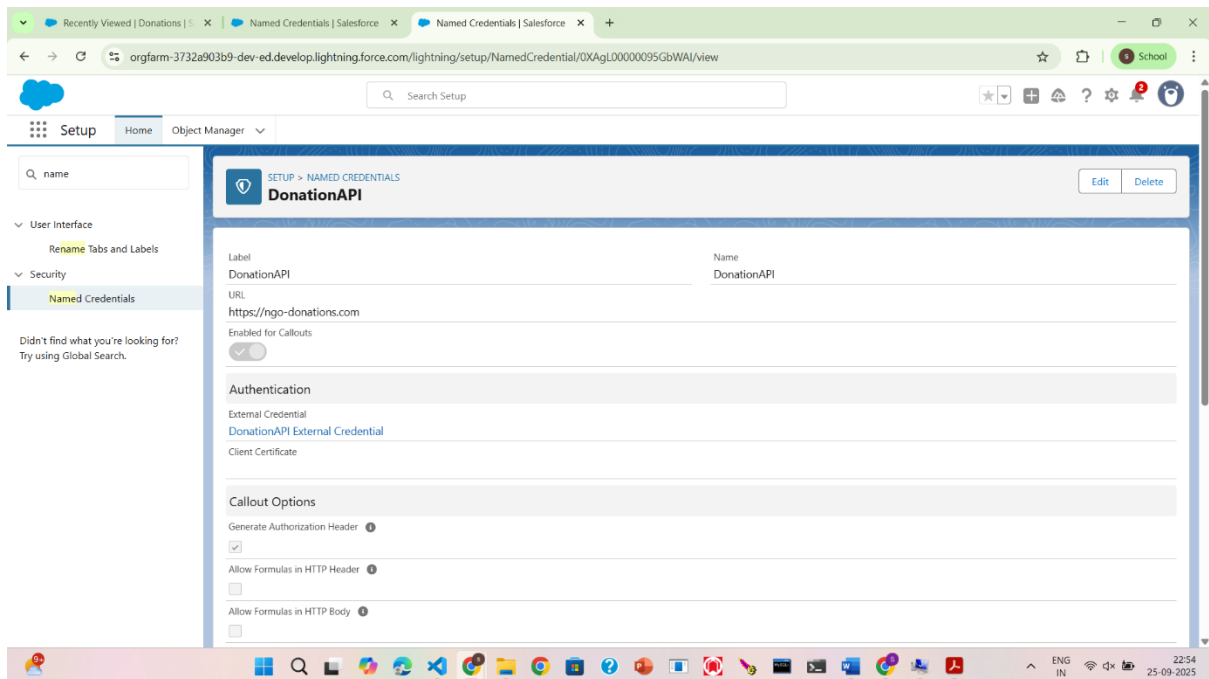
2. Fill in details:

- o Label: DonationAPI

- o URL: <https://ngo-donations.com>

- o External Credentials: DonationAPI External Credential.

3. Save → Use in Apex/Flows.



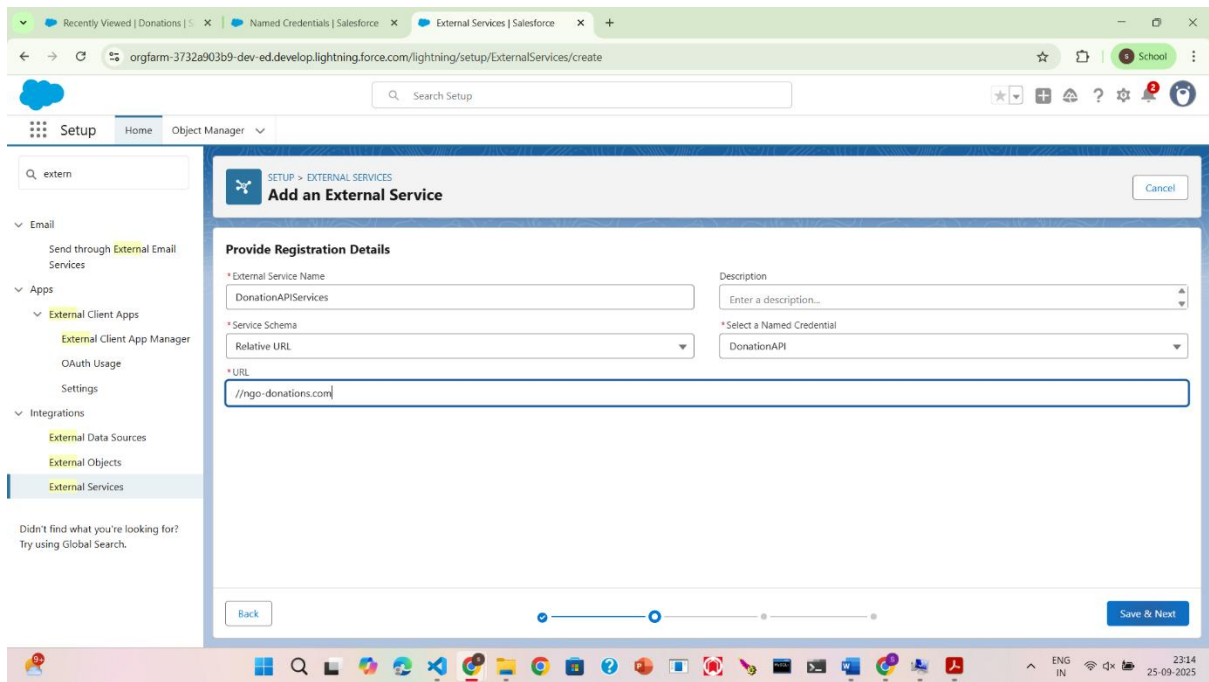
- **Step 2: External Services (No-code API integration)**

1. Go to Setup → External Services.

2. Click New External Service → upload API schema (Swagger file).

3. Salesforce auto-generates actions (like “Get Panel Output”).

4. Use these in Flows → e.g., when “Check Output” button is clicked, it fetches data from IoT.



• Step 3: Web Services (REST/SOAP)

- REST → You can call an external API to get donor or payment info from a payment gateway (e.g., Razorpay, Paypal, NGO backend).
- SOAP → If your NGO's backend or payment system exposes a WSDL, you can import it the same way.

REST Callout Example (Apex):

```
HttpRequest req = new HttpRequest ();
req.setEndpoint ('callout:DonationAPI/donations?donationId=123');
req.setMethod ('GET');
Http http = new Http();
HttpResponse res = http.send(req);
System.debug('External Donation API Response: ' + res.getBody());
```

• Step 4: Callouts (Triggered APIs)

Why? Automate sending donation data out when events happen.

Examples:

- When a new donation record is created, send it automatically to your NGO's backend system.

- When a donation is updated (status changed to “Received” or “Acknowledged”), trigger a REST call to your payment gateway or CRM.

- **Step 5: Platform Events (Real-time Alerts)**

Examples:

If a large Donation is made in Salesforce → publish DonationAlertEvent.

NGO Admin Dashboard subscribes → sends notification to the donation team.

How?

1. Go to Setup → Platform Events → New.

2. Create DonationAlertEvent with fields like:

DonationId (Text 18)

DonorName (Text 80)

DonationAmount (Number 16,2)

AlertType (Text 50)

The screenshot shows the Salesforce Setup interface for creating a new Platform Event. The left sidebar contains a navigation menu with categories like Events, Integrations, Privacy Center, Identity, and Security. The main content area is titled 'Platform Events' and contains a 'Platform Event Definition Edit' form. The form has sections for 'Platform Event Information' and 'Deployment Status'. In the 'Platform Event Information' section, the 'Label' is 'Donation Alert Event', the 'Plural Label' is 'Donation Alert Events', and the 'Object Name' is 'Donation_Alert_Event'. The 'Event Type' is set to 'High Volume' and the 'Publish Behavior' is 'Publish After Commit'. In the 'Deployment Status' section, the 'Deployed' radio button is selected. The bottom of the screen shows a Windows taskbar with various application icons and a system clock indicating 00:24 on 26-09-2025.

- **Step 6: Change Data Capture (CDC) for Donation Tracking**

Example:

If Donation Amount or Donation Status changes → Notify external NGO finance system.

If Donor Contact Info is updated → Notify donor management system.

How to Set It Up

1. Go to Setup → Change Data Capture

In Quick Find type “Change Data Capture

Click it.

2. Enable CDC on Your Objects

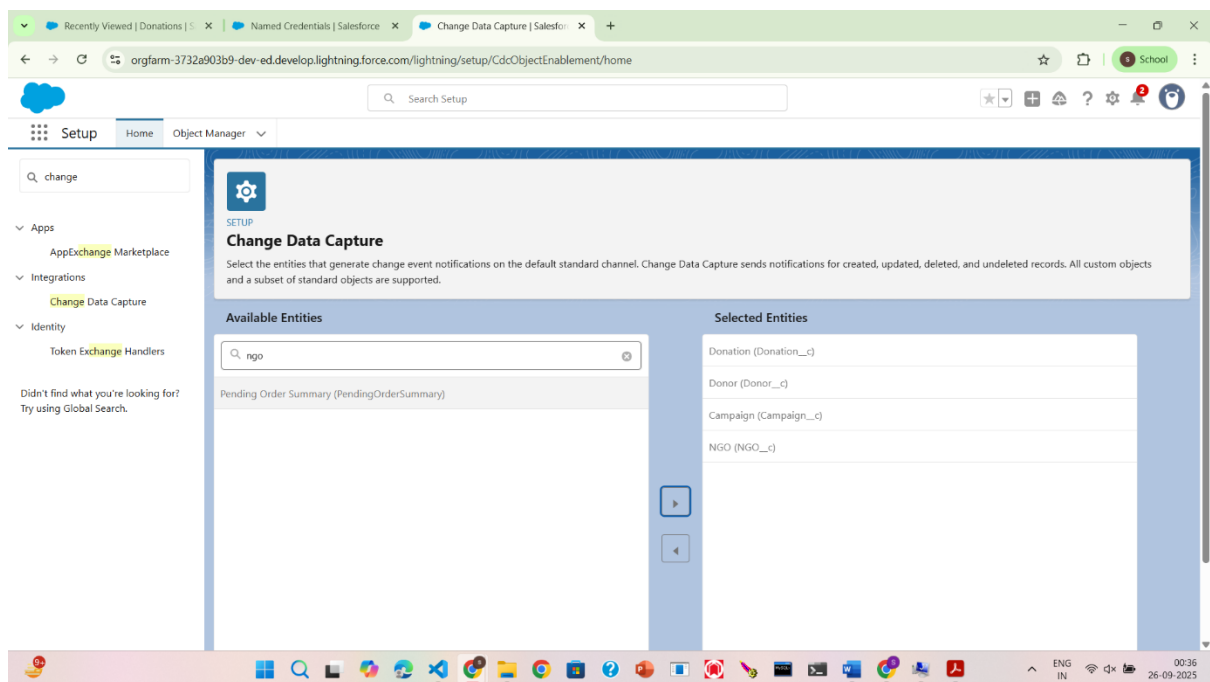
Select your objects like:

Donation__c (custom donation object)

Donor__c (donor object, if you have it)

Any related objects you want (like Campaigns, Projects etc.)

Click Save.



- **Step 7: Salesforce Connect (View external data in Salesforce)**

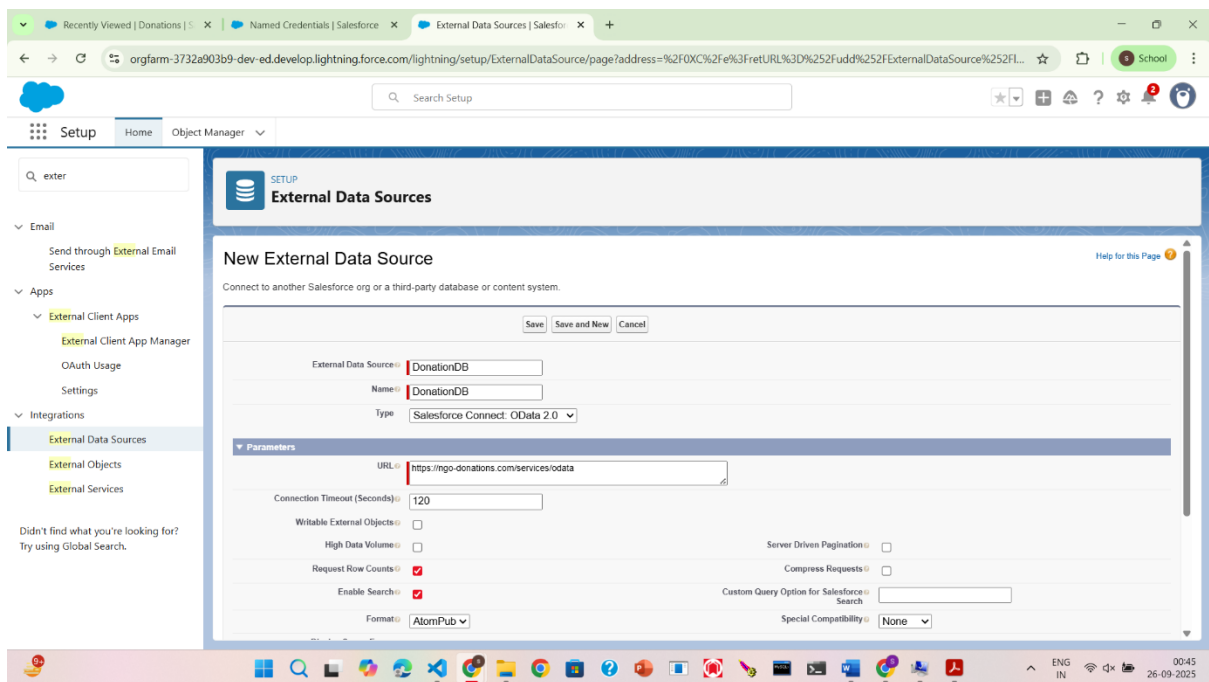
Example: Show donation records or donor payment logs stored in an external SQL database (or payment gateway) as if they're inside Salesforce.

How?

1. Setup → Salesforce Connect.

2. Create External Data Source → connect to database.

3. Panels appear as External Objects → viewable like normal records.



- **Step 8: API Limits (Donation Data Updates)**

Why? Salesforce has daily API call limits.

Tips:

If the external donation system sends thousands of donation updates → use Bulk API or Streaming API instead of individual REST calls.

Monitor usage in Setup → System Overview → API Usage to make sure the NGO's Salesforce org does not exceed daily limits.

- **Step 9: OAuth & Authentication (NGO Donor Portal Login)**

Example: Donors log in to an NGO Portal to view their donation history or receipts.

How?

1. Setup → Identity Provider in Salesforce.
2. Enable OAuth Providers (Google, Microsoft, Facebook, etc.) so donors can sign in with existing accounts.
3. Link the donor portal login with OAuth to securely authenticate donors and allow them to view or manage their donations.

- **Step 10: Remote Site Settings (Allow Callouts to External Donation API)**

Example: Before calling <https://ngo-donations.com> from Salesforce Apex or Flows, add it to Remote Site Settings.

How?

1. Go to Setup → Remote Site Settings.
2. Click New Remote Site.
3. Fill in the details:

Name: DonationAPI

URL: <https://ngo-donations.com>

4. Click Save.

